M Mohaiminul Islam

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+8801304869568 HuggingFace LeetCode Kaggle

Education

South Asian University, New Delhi, India
Master of Science in Computer Science *Total CGPA: 6.24/9.0*

Noakhali Science and Technology University
Bachelor of Engineering in Computer Science and Telecommunication Engineering
Total CGPA: 3.14/4
Last 4 semesters: 3.34/4

Research Interests

Legal NLP and Text Analytics, Low-Resource Language Technologies, Natural Language Processing, Large Language Models, Retrieval-Augmented Generation (RAG), Knowledge Graphs, Knowledge-Based AI, Graph Neural Networks, Graph Representation Learning, Speech Processing, Multimodal AI, Vision-Language Models, Agentic AI, Social Network Analysis, and AI in Mental Health.

Thesis

BSc Thesis Project: Recognition and Classification of Reptile Species using Deep Learning Techniques

Supervisor: Koushik Chandra Howlader 2017-2018

- Developed a research project focused on leveraging deep learning techniques for the automatic identification and categorization of reptile species using visual data.

MSc Thesis: Learning Stress-Specific Embeddings using Contrastive Learning over Skip-Gram Model for Stress Detection from Social Media Posts.

Supervisor: Dr. Amit Banerjee 2020-2021

- Proposed a stress-detection approach for social-media text by learning stress-specific embeddings through a Skip-Gram formulation over Transformer-based representations with supervised contrastive learning. Evaluated the approach with BERT and MentalBERT across three datasets, demonstrating improved performance and cross-dataset generalizability.

Publications

Conference Paper (Best Paper Award)

M Mohaiminul Islam and Muhammad Abulaish. "IndicIPR-QA: A Benchmark for Extractive Question Answering on Indian IPR Laws". *Best Paper in the Application, Data, Benchmark Track; Paper 147; Long (Oral) Presentation at the 13th IEEE International Conference on Data Science and Advanced Analytics (DSAA 2026).*

Conference Paper (Accepted, Withdrawn)

M Mohaiminul Islam. "ALPS: Admissibility-Preserving Semantic Partitioning for Region-Based Parallel Mining". *Accepted to the International Conference on Database Systems for Advanced Applications (DASFAA 2024); subsequently withdrawn due to funding unavailability.*

Journal Manuscripts (Submitted)

M Mohaiminul Islam and Muhammad Abulaish. "Auditing Knowledge-Graph-Augmented Legal Question Answering: Conformance, Baseline Contamination, and Benchmark Headroom". *Submitted to Artificial Intelligence and Law.*

M Mohaiminul Islam and Muhammad Abulaish. "Beyond Answer Extraction: Diagnosing Retrieval and Source Selection Failures in Indian Statutory Question Answering". *Submitted to Artificial Intelligence and Law.*

M Mohaiminul Islam and Muhammad Abulaish. "Domain-Adaptive Pretraining for Statutory Question Answering: Evidence from Authentic, Synthetic, and Mixed Legal Text". *Submitted to Natural Language Processing, published by Cambridge University Press.*

M Mohaiminul Islam, Roksana Akter, Diya Rahman, Abdullah Al Noman, Tze Hui Liew, Dip Nandi, Mashiour Rahman, and Abdullah Al Jubair. "Automatic Speech Recognition in Bengali Low-Resource Language Dialects: A Mini Review of Applications, Challenges, and Emerging Trends". *Submitted to Frontiers in Communication.*

M Mohaiminul Islam, Roksana Akter, Diya Rahman, Abdullah Al Noman, Tze Hui Liew, Dip Nandi, Mashiour Rahman, and Abdullah Al Jubair. "BanglaMix: A Multi-Dialect Bengali Speech Dataset and Benchmark for Code-Switched ASR". *Submitted to Language Resources and Evaluation.*

Projects

A Multi-Label NLP Classifier for Fine-Grained Quotation Categorization | *ONNX, Transformers* — April 2023
- Developed a multi-label NLP classifier to categorize quotations across 138 fine-grained categories. **Link**

Deep Learning-Based Vehicles Image Recognizer | *PyTorch, fastai, ETL, Hugging Face, Google Colab* — Dec 2023
- Developed a deep-learning image classifier for recognizing multiple vehicle categories. **Link**

Enhancing Waste Management through Computer Vision | *YOLOv5, DVC, MLflow, CI/CD, AWS* — April 2023
- Developed a computer-vision pipeline for waste detection to support automated environmental monitoring. **Link**

Prison Management System | *PHP, HTML/CSS, Node.js, MySQL, Git, Unix Shell, VS Code* — May 2017
- Developed a desktop application for managing prison details, including prisoner addition, updating, and searching. **Link**

Cryptocurrency Market Analysis | *Python, Selenium, Tableau* — Oct 2023
- Built a data collection and Tableau analysis workflow to explore cryptocurrency market trends. **Link**

Basketball Data Graph Analysis | *Neo4j, Cypher* — Jan 2025
- Modeled basketball data as a Neo4j graph and used Cypher queries to analyze relationships and extract graph-based insights. **Link**

Standardized Test Scores

IELTS — Overall Band Score: 6.5 (L: 7, R: 6, W: 6, S: 6.5)
Date Taken: 7 October 2023

Skills

Languages: Python, C, C++, Java

Analysis Tools: Neo4j, GraphViz, Gephi, Matlab, LLVM (Instrumentation Tool)

Ontology Tools: Protégé

Machine Learning Packages & Technologies: PyTorch, NumPy, Pandas, LangChain, LangGraph, Keras, NLP packages (e.g., BERT, Gensim, NLTK, spaCy), Scikit-learn, Transformers, Embedding techniques (e.g., Word2Vec, FastText, GloVe), Seaborn, BeautifulSoup, Selenium.

Big Data Frameworks: Hadoop MapReduce, Apache Spark, Cassandra, Elasticsearch

Development & Software Frameworks: Android, Docker, Git, CI/CD, MLflow, Django, Flask, FastAPI

Databases: Oracle, MySQL, CQL, PostgreSQL

Scripting: LaTeX, HTML, Prolog

Research, Teaching, and Professional Experience

Database Management Lab, South Asian University | *MySQL, PL/SQL, MongoDB*
- Worked on relational and NoSQL databases including schema design, normalization, query optimization, triggers, and aggregation pipelines.

Data Mining Lab, South Asian University | *Python, Weka, Scikit-learn*
- Implemented data preprocessing, classification (SVM, k-NN, Decision Trees), clustering (K-Means, DBSCAN), text mining, web mining, and dimensionality reduction (PCA).

Network Science Lab, South Asian University | *Neo4j, Cypher*
- Taught graph theory concepts including centrality measures, community detection, network partitioning, DAGs, and shortest path algorithms using Neo4j.

Elite Lab AI | *Research Assistant*
- Conduct research on low-resource language technologies, focusing on Bengali dialects and code-switched Bengali–English speech using NLP and Automatic Speech Recognition (ASR).

MasterCourse (EdTech) | *Data Science Instructor*
- Taught Data Science and supervised projects in Computer Vision, Data Analytics, Tableau dashboards, and NLP applications.

Ahom Limited | *Blockchain Developer*
- Worked on blockchain systems focusing on consensus efficiency and transaction optimization.

Honors and Awards

Merit Scholarship at South Asian University for the fall, winter, and spring quarters.
Second runner-up in the regional Math Olympiad in 2012.
First runner-up in an intra-school debate competition.